

The developers' viewpoint

Although many development environments exist for Android, the default way is to use Android SDK or Android Studio. For this, you have to:

- Learn Java
- Install Android Studio/SDK
- Get familiar with the Android API
- Compile each time to test even minor changes

But the Firefox OS has an innovative way of app development compared to Android. Firefox apps are basically Web apps and this makes the development highly clutter-free. Let us have a look at it.

Firefox OS: Something different

If you thought that an OS that can run just one app would be of no use, Mozilla proves you wrong. Technically speaking, the Firefox OS is simply the Firefox browser running on top of the Linux kernel. User apps that you choose from the menu are just Web pages (or Web apps - no matter if offline or online), rendered by the Firefox browser. But from a user's viewpoint, everything happens just like in Android. One never knows that the apps are just some Web pages rendered by a Web browser engine. Clever, isn't it?

As the Web browser and Internet technologies evolved, the idea of the Web browser as a universal application was developed. Everything you need happens inside the browser window. Webmail services like Gmail proved this is possible. There are a lot of websites out there which act like small image editing programs or office suites. We are familiar with cloud services like Google Docs. And finally, the rise of HTML5 is giving a big boost to Web apps.

Firefox OS began as a project codenamed 'Boot to Gecko', which means you are entering the Gecko HTML rendering engine (the engine that powers the Firefox browser), which will act as an operating system. Execution of apps will be the rendering of HTML pages and corresponding scripts.

The question is, why should such a system exist? It is clear that native applications (compiled binaries) will have performance advantages compared to interpreted ones like Web apps. Let us check if there is something to balance this performance issue.

Android development vs Firefox OS development

We have had an overview of Android development in a previous section. One creates apps using Java and the Android SDK, and distribute them after compilation.

Although compiled apps can achieve higher performance, Web apps like the ones developed for Firefox OS have the following advantages:

- Easy to develop

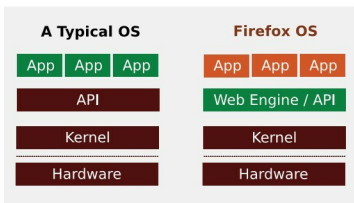


Figure 2: Comparison of the basic structure of a typical OS and the Firefox OS

- Easy to maintain
- Portable across devices
- Portable across OSs

Let us break the 'Easy to develop' thing apart. Web apps are written using HTML, CSS and JavaScript. This means you are not forced to learn something new that has limited scope. You are not required to install and use a proprietary SDK. When you are learning HTML and JavaScript, you are gaining some skill useful in many other fields. Also, you can practice Web app development using any good Web browser, in any platform. In case of the Firefox OS, the Firefox browser and some add-ons will be sufficient. There's no need for gigabytes of SDK downloads.

Web apps are easy to maintain since they are not bound to any specific SDK. Also, there is no need to build a package before you test minor changes. There will be a few compatibility issues since Web apps are based on standards like HTML5, which also makes them portable across devices and browsers.

Undoubtedly, Android is a good operating system. It has a friendly interface, a large number of supported apps, and so on. But its success is not a product of its quality alone. The alliance of handset manufacturers contributes to its enjoying a near-monopoly. Also, it began some years before Firefox OS or Ubuntu Touch, which means it has had enough time to impress us. However, we are happy about the success of Android since it is mostly open source.

We discussed how Firefox OS has pioneered the concept of implementing a Web app strategy for app development. Android still has to do something like this in order to become more developer-friendly.

Let us hope that collaboration between these systems will help all the free software mobile OSs to become more powerful and useful. 

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